The Business Problem

Hotel Booking Cancellation Prediction

- Hotels lose revenue when customers cancel reservations.
- Last-minute cancellations make room planning difficult.
- Empty rooms can lead to lost revenue and inefficient staffing.
- Hotels need a way to identify bookings that are more likely to be cancelled.

Business objective: Predict booking cancellations early so hotels can make better business decisions.

The Data

understanding Hotel Booking Data

- **Source:** Hotel booking demand dataset.
- **Dataset size:** Approximately 119,390 booking records before preprocessing.
- **Target variable:** `is_canceled`
  - 0 = Booking not cancelled
  - 1 = Booking cancelled
- **Key features:** Lead time, arrival date, deposit type, customer type, market segment and previous cancellations.
- **Data-quality issues:** Missing values, categorical variables and potential data leakage.

Business value: Historical booking data helps identify patterns associated with cancellations.

The Approach

How Machine Learning Solves the Problem

- **Problem type:** Binary classification.
- **Models compared:** Logistic Regression, Decision Tree and Random Forest.
- **Data preparation:** Handled missing values, encoded categorical features and scaled numerical features where required.
- **Evaluation:** Compared models using the F1 Score on unseen test data.
- **Why F1 Score?** It balances precision and recall, helping evaluate cancellation predictions.

Business value: Compare different models and select the one that best balances identifying cancellations with avoiding false alarms.

Model	Training Accuracy	Testing Accuracy	Test F1 Score
Logistic Regression	73.73%	73.85%	58.58%
Decision Tree	79.01%	78.70%	69.87%
Random Forest	83.81%	82.43%	73.72%

Key Findings

- Logistic Regression achieved a Test F1 Score of 58.58%.
- Decision Tree improved the Test F1 Score to 69.87%.
- Random Forest achieved the highest Test F1 Score of 73.72%.
- Random Forest also achieved the highest testing accuracy of 82.43%.

Recommendation and Next Steps

Turning Predictions into Business Value

- Pilot the model to identify bookings with a higher predicted cancellation risk.
- Send reminders or offer flexible booking options where appropriate.
- Use predictions to support room planning and staffing decisions.
- Monitor precision, recall and F1 Score on new bookings.
- Assess financial impact before introducing operational changes.

Risks and limitations:

- Predictions are not guarantees that a customer will cancel.
- Historical patterns may change over time.
- False predictions could lead to unnecessary costs or poor customer experiences.
- Financial benefits have not yet been measured.

Final recommendation: Run a controlled pilot, measure the results and refine the model before wider implementation.